Chromium Browser Artifact Parser

User's Guide

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Repository: https://github.com/jjrboucher/Chromium-Browser-Artifact-Parser

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1. Introduction

The Chromium Browser Artifact Parser is a forensic tool designed to extract and parse browser artifacts from Chromium-based web browsers. It provides a graphical user interface (GUI) that simplifies the process of extracting valuable forensic data from browser profiles and exports the results to a well-organized Microsoft Excel spreadsheet.

Key Features

- Modern GUI Interface — User-friendly graphical interface with progress tracking and status updates
- Selective Artifact Processing — Choose which artifacts to extract based on your investigation needs
- Comprehensive Data Extraction — Parses 22 different artifact categories from browser profiles
- Multi-Browser Support — Works with Chrome, Microsoft Edge, Opera, and other Chromium-based browsers
- Gap Detection — Identifies deleted records by detecting gaps in database record numbering
- Timestamp Decoding — Automatically converts WebKit timestamps to human-readable UTC format
- History Clusters Analysis — Parses Chrome's browsing session clusters with search terms, content annotations, and engagement scores
- Extensions Inventory — Extracts installed extension metadata from each extension's manifest.json
- Excel Output — Organizes all extracted data into a multi-worksheet Excel file with a summary sheet

Use Cases

- Digital forensic investigations
- Incident response
- Employee monitoring (with proper authorization)
- Personal data recovery
- Security audits

2. System Requirements

Hardware Requirements

- Minimum 4 GB RAM (8 GB recommended)
- 100 MB free disk space for the application
- Additional disk space for output files (varies based on browser data volume)

Software Requirements

- Operating System: Windows 10/11 (primary), also works on macOS and Linux
- Python: Version 3.10 or higher

Required Python Packages:

- pandas ~= 2.0.1
- openpyxl ~= 3.1.5
- numpy ~= 1.24.3
- DateTime ~= 5.5
- tkinter (typically included with Python)

Browser Compatibility

- Google Chrome (tested with versions 129 and 145)
- Microsoft Edge (tested with version 129)
- Opera (tested with version 113)
- Other Chromium-based browsers (Brave, Vivaldi, etc.)

Note: The tool is designed for desktop browser profiles. Mobile browser profiles may have different database structures and are not currently supported.

3. Installation

Option 1: Running from Source (Recommended)

1. Clone or download the repository from GitHub:

   ```
   https://github.com/jjrboucher/Chromium-Browser-Artifact-Parser
   ```

2. Install Python dependencies from the project directory:

   ```
   pip install -r requirements.txt
   ```

3. Run the application:

   ```
   python browser-artifact-parser-GUI.py
   ```

Option 2: Standalone Executable (Windows Only)

Download browser-artifact-parser-GUI.exe from the repository and double-click to run — no Python installation required.

4. Getting Started

Locating Browser Profile Folders

Before using the parser, locate the browser profile folder you want to analyze. Default locations:

Browser	Windows Path
Google Chrome	%LOCALAPPDATA%\Google\Chrome\User Data\Default
Microsoft Edge	%LOCALAPPDATA%\Microsoft\Edge\User Data\Default
Opera	%APPDATA%\Opera Software\Opera Stable
Brave	%LOCALAPPDATA%\BraveSoftware\Brave-Browser\User Data\Default
Vivaldi	%LOCALAPPDATA%\Vivaldi\User Data\Default

```
macOS Chrome: /Users/<username>/Library/Application Support/Google/Chrome/Default
Linux Chrome: /home/<username>/.config/google-chrome/Default
```

Note: If the user has signed into the browser, profiles may be named Profile 1, Profile 2, etc., instead of Default.

Important Considerations

- Close the Browser First — Browser databases are locked while the browser is running. Close all browser windows before parsing.
- Make a Copy — For forensic investigations, always work with a copy of the profile folder, not the original.
- Administrator Access — You may need administrator privileges to access profile folders in protected locations.

5. Using the Application

Main Interface Overview

- Chrome User Profile Folder — Path to the browser profile to analyze
- Output Excel File — Path where the results will be saved
- Artifact Selection — Checkboxes to select which artifacts to process
- Progress Section — Progress bar and control buttons
- Processing Status — Real-time log of processing activities

Step-by-Step Instructions

Step 1: Select Browser Profile Folder

- Click the Browse button next to 'Chrome User Profile Folder'
- Navigate to and select the browser profile folder (e.g., Default)
- The path will appear in the text field

Step 2: Select Output File Location

- Click the Save As button next to 'Output Excel File'
- Choose a location and filename for the Excel output
- The file will automatically have the .xlsx extension

Step 3: Select Artifacts to Process

Use the checkboxes to select which artifacts to extract. You can use Select All, Deselect All, or manually check/uncheck individual artifacts.

Category	Artifact	Description
History	History	Complete browsing history
	History Gaps	Gaps in history record numbering (potential deletions)
	History Clusters	Browsing session clusters with content annotations and search terms (6 worksheets)
Downloads	Downloads	Downloaded file information
	Downloads Gaps	Gaps in download record numbering
Form Data	Autofill	Form field autofill entries
	Addresses	Saved address information
	Credit Cards	Masked credit card information
	Bank Accounts	Masked bank account information with usage data
	Loyalty Cards	Store loyalty/rewards card information
Login	Login Data	Saved login credentials (passwords encrypted)
	Login Data Gaps	Gaps in login record numbering
Search	Search Terms	Search queries and associated URLs
	Keywords	Search engine keyword information
Navigation	Shortcuts	Omnibox shortcuts and visit counts
	Top Sites	Most visited websites

Category	Artifact	Description
Bookmarks	Bookmarks	Saved bookmarks (including backup)
Cookies	Cookies	Cookie data and timestamps
Media	FavIcons	Website favicons and metadata
Settings	Preferences	User preferences and account information
Extensions	Extensions	Installed extension inventory from manifest.json files
Edge Only	Web Assist (Edge)	Edge-specific navigation history

Step 4: Run the Parser

- Click the Run Parser button
- Monitor progress in the status window
- Wait for the 'Processing completed successfully!' message

Step 5: Access Results

When processing completes, a success dialog will display the output file location. Open the Excel file to review the extracted data.

Stopping Processing

Click the Stop button to halt processing before completion. Partial results will be available in the output file.

6. Understanding the Output

Excel File Structure

- Summary (first sheet) — Record counts for each artifact category, plus script version
- Preferences (second sheet) — Parsed user preferences and account info
- Individual artifact sheets — One or more sheets per processed artifact

Note: The History Clusters artifact produces six separate worksheets rather than one.

Timestamp Columns

- Raw timestamps — Original WebKit format (microseconds since January 1, 1601)
- Decoded timestamps — Human-readable UTC format (e.g., '2024-03-15 14:32:45')

7. Supported Browsers

Google Chrome

Full support for all artifacts. This is the primary browser the tool was developed and tested against (versions 129 and 145).

Microsoft Edge

Full support with additional Edge-specific features. The tool automatically detects Edge profiles by checking for 'edge' in the path, parses the WebAssistDatabase unique to Edge, and extracts Edge-specific account information from Preferences.

Opera

Tested and working with Opera 113. Same artifact support as Chrome.

Other Chromium Browsers

The tool should work with Brave, Vivaldi, Chromium, and other Chromium-based browsers. However, unique database tables or fields in those browsers may not be parsed.

8. Artifact Reference

History Artifacts

History

Extracted from the History SQLite database. Includes transition types, visit qualifiers, visit duration, and URL segments.

History Gaps

Detects deleted history records by finding gaps in record ID sequences. Results include previous and current record numbers, count of missing records, and timestamp boundaries.

History Clusters

New in version 2026-Mar-1. Parses Chrome's session clustering tables from the History database. When selected, this artifact produces six separate worksheets:

Worksheet	Description
Clusters Overview	One row per cluster with label, keywords (where available), prominent flag, and total visit count
Clusters Contents	Every visit within each cluster: timestamps, URLs, titles, relevance scores, engagement scores, visit duration, and transition details
Clusters Search Term	Search terms associated with cluster visits, derived from the content_annotations table. Includes temporal span of the search activity
Clusters Timeline	First/last visit timestamps, session duration, visit count, and unique URL count per cluster
Clusters Duplicate Visits	Pairs of visits that Chrome's engine identified as duplicates, with timestamps and time gap between them
Clusters Comprehensive Export	Full join of cluster data with content_annotations (search terms, page categories, entities) and context_annotations (foreground duration, page end reason)

Important: The cluster_keywords table is empty in current Chrome builds (the Journeys UI was removed in Chrome 125, May 2024). The Clusters Search Term worksheet derives search term data from the content_annotations table instead.

Downloads

Extracted from the downloads and downloads_url_chains tables in the History database. Includes file state, danger type, interrupt reason, and download source URL.

Form Data Artifacts

Autofill

Form field data users have entered, including field name and value, usage count, and creation/last used dates.

Addresses

Saved address book entries with name, email, phone, street address, city, state, postal code, country, and company.

Credit Cards

Masked credit card information (last 4 digits only), including card network, expiration date, billing address reference, and usage stats.

Bank Accounts

Masked bank account information including bank name, account type, account number suffix, nickname, usage count, and last use date.

Loyalty Cards

Store loyalty and rewards card information including merchant name, domain, program name, logo URL, and loyalty card number.

Extensions

New in version 2026-Mar-1. Reads each extension's manifest.json from the Extensions/ subfolder of the profile. Columns: ID, Name, Version, Description, Author, Homepage URL.

Preferences Artifact

Parsed from the JSON Preferences file. Includes account information (email, full name, GAIA ID), profile creation details, download directory, startup pages, homepage, exit status, and new tab shortcuts.

9. Troubleshooting

'Database is locked' Error

Cause: The browser is still running and has locked the database files. Close all browser windows and check Task Manager for remaining browser processes before retrying.

'No such file or directory' Error

Cause: The selected folder is not a valid browser profile or is missing expected files. Verify you selected the profile folder itself (containing files like History, Cookies, Preferences), not the parent 'User Data' folder.

Empty or Missing Worksheets

Cause: The corresponding database table may be empty or the browser version uses a different structure. Check the Summary worksheet for '0 records' entries. Some artifacts (like Bank Accounts or History Clusters) may legitimately be empty.

History Clusters Worksheet Is Empty

Cluster tables exist in the schema of all modern Chrome builds but may contain no data if the user has not browsed enough to trigger cluster formation, or if Chrome has not yet written cluster data for recent sessions.

'Module not found' Error

Run: pip install -r requirements.txt

Performance Issues

For profiles with extensive history, select only the artifacts you need and ensure adequate disk space for the output file. History-heavy profiles can generate large Excel files that may be slow to open.

10. Frequently Asked Questions

Q: Can this tool decrypt saved passwords?

A: No. The tool extracts encrypted password data, but decryption requires the user's Windows credentials and is not implemented.

Q: Does this tool modify the original browser data?

A: No. The tool opens databases in read-only (immutable) mode and never writes to or modifies the browser profile.

Q: What is the History Clusters artifact?

A: Chrome groups related browsing sessions into 'clusters' based on topic similarity. The History Clusters artifact parses these cluster tables to reveal research sessions, associated search terms, page engagement scores, and duplicate visit patterns. It produces six separate worksheets.

Q: Why is the Clusters Search Term worksheet empty?

A: The cluster_keywords table is not populated in current Chrome builds (the Journeys feature was removed in Chrome 125). Search terms are instead derived from the content_annotations table. If that table also contains no data, the worksheet will be empty.

Q: What does the Extensions artifact capture?

A: It scans the Extensions/ folder inside the profile, reads each extension's manifest.json, and extracts the extension ID, name, version, description, author, and homepage URL.

Q: What's the difference between History and History Gaps?

A: 'History' extracts all browsing records. 'History Gaps' identifies where records have been deleted by finding missing sequence numbers in the database.

Q: Why is 'Web Assist (Edge)' disabled by default?

A: This artifact is specific to Microsoft Edge and doesn't exist in Chrome or other browsers. Enable it only when analyzing Edge profiles.

Q: How far back does browser history go?

A: Chrome retains approximately 90 days of browsing history. The History Clusters artifact may contain data from within that same window, derived from the same History database.

Q: What timezone are the decoded timestamps in?

A: All decoded timestamps are in UTC (Coordinated Universal Time). Convert to local time as needed for your investigation.

Appendix A: File Locations Quick Reference

Browser	Windows Path
Chrome	%LOCALAPPDATA%\Google\Chrome\User Data\Default
Edge	%LOCALAPPDATA%\Microsoft\Edge\User Data\Default
Opera	%APPDATA%\Opera Software\Opera Stable
Brave	%LOCALAPPDATA%\BraveSoftware\Brave-Browser\User Data\Default
Vivaldi	%LOCALAPPDATA%\Vivaldi\User Data\Default

Appendix B: Transition Type Reference

Value	Type	Description
0	LINK	Clicked a link
1	TYPED	Typed URL in address bar
2	AUTO_BOOKMARK	Clicked suggestion in UI
3	AUTO_SUBFRAME	Automatic subframe navigation
4	MANUAL_SUBFRAME	User-initiated subframe navigation
5	GENERATED	Typed text that didn't look like URL
6	START_PAGE	Top-level navigation
7	FORM_SUBMIT	Submitted form data
8	RELOAD	Reloaded the page
9	KEYWORD	Keyword-generated URL
10	KEYWORD_GENERATED	Visit generated for keyword